

Alex Lee

alex.lee.swe@gmail.com | +1 778-512-6634

 github |  linkedin

SKILLS

Languages/ Databases Javascript | Python | TypeScript | R | Java | C | C++ | HTML | CSS | SQL | NoSQL
Technology React.js | Next.js | FastAPI | Flask | Tensorflow | Pytorch | Scikit-learn | Spring | Docker

EXPERIENCES

Boeing

Jan 2024 - Aug 2024

Software Developer Intern

Richmond, BC

- Contributed to Insight Accelerator, a visualization and analytics tool for airlines, as a full-stack developer using React, Python, HTML, and Docker
- Enhanced the tool's functionality by implementing interactive graphs and extending analytical features, improving data visualization and analysis
- Identified and optimized code areas to significantly enhance frontend responsiveness by improving backend and frontend interactions, utilizing techniques like optimistic updates
- Collaborated in an agile environment working closely with developers, product managers, and UI designers to deliver high-quality features and improvements

BRP

Jan 2023 - Aug 2023

Software Developer Intern

Valcourt, QC

- Developed CI/CD pipelines and Python automation scripts on Azure DevOps to manage internal accounts, including auditing and removing unused Jira licenses to reduce operational costs
- Worked with Microsoft Power Platform (Power Apps, Power Automate, Power BI, Power Virtual Agents) to build internal tools and automate business workflows on Azure DevOps

NOTABLE PROJECTS

TradingCast

- Engineered a full-stack stock analysis platform using Next.js and FastAPI featuring candlestick chart visualization with technical indicators, ML-based price prediction, and stock correlation analysis for portfolio diversification.
- Trained a PyTorch deep learning model with embeddings and scikit-learn pipeline to forecast stock price movements, achieving 40-50% directional accuracy against a 33% random baseline using multi-stock correlation as training input
- Implemented a stock correlation feature using Python and pandas to quantify price movement relationships across multiple stocks, and a CSV export pipeline enabling users to download historical data for offline analysis

Agrobot UBC

- Collaborated with a team of 80 developers to build a fully autonomous robot capable of analyzing its environment to perform targeted weeding, fertilizing, and data collection
- Contributed to the Image Recognition subteam, leveraging algorithms and technologies such as TensorFlow's object identification module to distinguish between plants and objects in real time

Animal Classifier

- Built a deep learning model to classify animal images (via link or upload) into one of 91 categories
- Achieved 95.97% accuracy using techniques such as transfer learning, fine-tuning, and data augmentation
- Developed a full-stack application utilizing React for the client side and Flask for server-side API endpoints hosting the TensorFlow-trained model

Food Order Website

- Developed a full-stack food delivery website using the MERN stack and REST API, enabling effective communication between customers and delivery staff
- Implemented security with JSON Web Token (JWT) authorization and password encryption using bcrypt
- Integrated automated email scheduling to notify customers and employees of order updates

EDUCATION

Bachelor of Science in Computer Science, Minor in Commerce

Sept 2020 - Apr 2025

University of British Columbia

Vancouver, BC